

THE ORDINARY CHOW RANK OF THE SIX-BY-SIX PERMANENT

ABSTRACT. Let perm_6 be the permanent of a generic 6 by 6 matrix. Over an algebraically closed field of characteristic zero, we prove that its ordinary Chow rank is 32. The upper bound is Glynn's identity. The lower bound combines a symmetric image-span inequality with a half-defect estimate for arbitrary quotient derivative symbols of a sextic Chow term. A proof audit exposed a false identification between formal subproduct spaces and actual derivative spaces in two dependent-factor profiles. This revision removes that identification, computes the actual spaces, and uses weaker sufficient local rows. The finite parts have an exact rational replay. No border-rank or general- n equality is claimed.

1. STATEMENT AND PERMANENT DERIVATIVES

Let $\mathbb{K}$ be algebraically closed of characteristic zero and let

$$V = \text{span}_{\mathbb{K}}\{x_{rc} : 1 \leq r, c \leq 6\}, \quad \dim V = 36.$$

For a homogeneous polynomial f , let $\mathcal{D}_j(f) \subset \text{Sym}^j V$ be the span of all derivatives whose remaining degree is j , and put $E_j = \mathcal{D}_j(\text{perm}_6)$.

Definition 1.1. A sextic Chow term is a nonzero product $T = \ell_1 \cdots \ell_6$. The ordinary Chow rank $R_{\text{Chow}, \mathbb{K}}(f)$ is the least number of such terms whose sum is f .

Theorem 1.2. Over $\mathbb{K}$,

$$R_{\text{Chow}, \mathbb{K}}(\text{perm}_6) = 32.$$

We first record the permanent derivative facts used below.

Lemma 1.3 (Permanent derivative tower). *One has*

$$\dim E_3 = 400, \quad \dim E_2 = 225, \quad \partial E_2 = V,$$

and

$$(1.1) \quad E_2^{(1)} := \{g \in \text{Sym}^3 V : \partial g \subset E_2\} = E_3.$$

Moreover, every nonzero element of E_2 has quadratic matrix rank at least four, and every nonzero element of E_3 has at least nine essential variables.

Proof. The 3×3 and 2×2 subpermanents are bases of E_3 and E_2 : distinct row-column blocks have disjoint matching supports. Their numbers are $\binom{6}{3}^2 = 400$ and $\binom{6}{2}^2 = 225$. Differentiating a rectangle permanent at one corner gives the opposite variable, proving $\partial E_2 = V$.

The inclusion $E_3 \subset E_2^{(1)}$ is immediate. Conversely, suppose all first derivatives of a cubic g lie in E_2 . Row and column multiaffinity forces every monomial of g to be a three-edge matching. Inside a fixed 3×3 block, the rectangle relations equate coefficients of matching monomials related by a transposition. Transpositions connect all six matchings, so each block is a scalar subpermanent. This proves (1.1).

The row-column diagonal torus has distinct characters on each subpermanent basis. A generic one-parameter subgroup degenerates any nonzero element to a single basis element, while matrix and first-catalectic ranks cannot increase under specialization. A rectangle permanent has quadratic

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rank four. The nine first derivatives of a 3×3 permanent are independent. The two rank floors follow. $\square$

2. SMALL PERMANENT-QUADRATIC INTERSECTIONS

Lemma 2.1. *If $L \subset V$ has dimension at most six, then*

$$\dim(E_2 \cap \text{Sym}^2 L) \leq 3.$$

If $\dim L \leq 5$, the upper bound improves to one.

Proof. For fixed a, q , the locus in $\text{Gr}(a, V)$ on which $\dim(E_2 \cap \text{Sym}^2 L) \geq q$ is closed, projective, and stable under the connected row-column torus. If nonempty, the Borel fixed-point theorem gives a torus-fixed point. Since the variable weight spaces are one-dimensional, such a point is a coordinate a -plane, equivalently a bipartite graph with a edges.

At a coordinate point, the intersection has one independent rectangle permanent for each four-cycle. Two distinct four-cycles have a union of at least six edges. If the union has exactly six edges, it is $K_{2,3}$ or $K_{3,2}$, which has exactly three four-cycles. Thus five edges support at most one four-cycle and six edges support at most three. $\square$

3. LOCAL LINEAR ALGEBRA

Polarization gives an injective map

$$\delta : \text{Sym}^3 L \longrightarrow L \otimes \text{Sym}^2 L.$$

Lemma 3.1 (Kernel-preimage bound). *Let $P : L \twoheadrightarrow D$ have rank d , let $S = \ker P$, and let $R \subset \text{Sym}^2 L$. Then*

$$\ker((P \otimes 1)\delta) = \text{Sym}^3 S,$$

and the inverse image of $D \otimes R$ has dimension at most

$$(3.1) \quad \boxed{\binom{\dim L - d + 2}{3} + d \dim R.}$$

Proof. The first equality follows in coordinates: all derivatives in directions modulo S vanish precisely when the cubic depends only on S . The inverse image of a subspace has dimension at most the kernel dimension plus the subspace dimension, which gives (3.1). $\square$

Lemma 3.2 (Squarefree projected-symbol table). *Let $\widehat{L} = \mathbb{K}^6$ with basis $z_1, \dots, z_6$, and let $\widehat{F}, \widehat{U}$ be the squarefree quadratic and cubic spaces. Set*

$$\delta(z_a z_b z_c) = z_a \otimes z_b z_c + z_b \otimes z_a z_c + z_c \otimes z_a z_b.$$

If $P : \widehat{L} \twoheadrightarrow D$ has rank d and $R \subset \widehat{F}$ has dimension at most $r \leq 3$, then the following table is a rowwise lower bound for $\text{rank}((P \otimes q_R)\delta)$:

$$(3.2) \quad \begin{array}{c|cccccc} r \backslash d & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 0 & 0 & 10 & 16 & 19 & 20 & 20 & 20 \\ 1 & 0 & 9 & 14 & 16 & 16 & 20 & 20 \\ 2 & 0 & 8 & 12 & 13 & 16 & 19 & 20 \\ 3 & 0 & 7 & 10 & 10 & 15 & 17 & 19. \end{array}$$

Proof. The diagonal torus preserves δ . The orbit closure of $(\ker P, R)$ in the relevant product of Grassmannians contains a fixed point, and rank can only drop in specialization. Thus $\ker P$ may be taken as a vertex set B , and R as the span of an edge set A with at most r edges. Put $O = [6] \setminus B$.

Distinct source cubics have disjoint output supports. The number of killed cubics is

$$(3.3) \quad \binom{|B|}{3} + |O|e_A(B) + \sum_{b \in B} \binom{\deg_A(b, O)}{2} + \#\{\text{triangles of } A[O]\}.$$

Inspection of edge sets of size at most three gives (3.2). The exact replay independently checks all 45,696 coordinate pairs and records a minimizer for each entry. $\square$

4. THE HALF-DEFECT ESTIMATE

Let $T = \ell_1 \cdots \ell_6 \neq 0$, and set

$$L = \text{span}\{\ell_1, \dots, \ell_6\}, \quad U = \mathcal{D}_3(T), \quad u = \dim U,$$

$$F = \mathcal{D}_2(T), \quad R = F \cap E_2.$$

For a quotient $P : L \rightarrow D$ of rank d , define

$$\beta_{P,R} : U \xrightarrow{\delta} L \otimes F \xrightarrow{P \otimes q_R} D \otimes (F/R).$$

Proposition 4.1 (Half-defect quotient-symbol estimate). *For every nonzero sextic Chow term and every quotient of its actual factor span,*

$$(4.1) \quad \boxed{\text{rank } \beta_{P,R} + \frac{20 - u}{2} \geq \frac{10}{3}d.}$$

Proof. Write $\ell = \dim L$.

Factor span at most four. Choose ℓ independent factors as coordinates. A generic diagonal one-parameter subgroup degenerates the remaining factors to unique coordinate initials, so T specializes to

$$x_1^{m_1} \cdots x_\ell^{m_\ell}, \quad m_j \geq 1, \quad \sum_j m_j = 6.$$

Middle rank cannot increase in the limit. Counting degree-three divisors over the positive partitions of six gives

$$(4.2) \quad \ell = 1, 2, 3, 4 \implies u \geq 1, 2, 4, 8.$$

Lemma 2.1 gives $\dim R \leq 1$. For $0 < d < \ell$, Lemma 3.1 gives

$$(4.3) \quad \text{rank } \beta_{P,R} \geq \max \left\{ 0, u - \binom{\ell - d + 2}{3} - d \right\}.$$

At $d = \ell$, a kernel cubic lies in $E_2^{(1)} = E_3$, but it is supported on at most six essential variables, contrary to Lemma 1.3. Thus the full symbol is injective. Minimizing over the conservative interval from (4.2) to $\min\{20, \binom{\ell+2}{3}\}$ gives

$$(4.4) \quad \begin{array}{c|c} \ell & \text{rank } \beta_{P,R} + (20 - u)/2 \text{ for } d = 0, \dots, \ell \\ \hline 4 & (0, 9/2, 8, 10, 14) \\ 3 & (5, 15/2, 9, 12) \\ 2 & (8, 9, 11) \\ 1 & (19/2, 21/2). \end{array}$$

Six independent factors. Here $u = 20$, the actual derivative spaces are the squarefree spaces, and $\dim R \leq 3$. The last row of (3.2), improved at $d = 6$ by full-symbol injectivity, gives

$$(4.5) \quad (0, 7, 10, 10, 15, 17, 20).$$

Five-dimensional factor span. The unique relation among the six factors puts T , after changing and rescaling coordinates, into exactly one form

$$(4.6) \quad T_s = x_1 x_2 x_3 x_4 x_5 (x_1 + \cdots + x_s), \quad 1 \leq s \leq 5.$$

Lemma 2.1 gives $\dim R \leq 1$. Exact differentiation of the actual polynomial gives

$$(4.7) \quad \begin{array}{c|ccccc} s & 1 & 2 & 3 & 4 & 5 \\ \hline \dim F & 11 & 11 & 13 & 14 & 15 \\ u & 14 & 14 & 18 & 20 & 20. \end{array}$$

The replay constructs the order-four and order-three derivative matrices of (4.6) in the ordinary monomial bases and row-reduces them over $\mathbb{Q}$.

For $s = 3$, the actual space U contains the nine squarefree cubics other than $x_1 x_2 x_3$. For $s = 4, 5$, it contains all ten squarefree cubics. These subspaces are stable under the diagonal torus. A nonzero rank-one quotient specializes to a coordinate functional, whose rank is the number of displayed triples through that coordinate: at least five for $s = 3$ and six for $s = 4, 5$. Quotienting by R loses at most one. Together with (3.1) and full-symbol injectivity this gives

$$(4.8) \quad \begin{array}{c|c|c} s & u & \text{rank } \beta_{P,R} + (20 - u)/2 \\ \hline 4, 5 & 20 & (0, 5, 8, 13, 15, 20) \\ 3 & 18 & (1, 5, 7, 12, 14, 19). \end{array}$$

For $s = 1$, the actual U is spanned by the ten squarefree cubics and $x_1^2 x_j$, $2 \leq j \leq 5$. If a direction $\lambda = \sum a_j \partial_{x_j}$ has $a_j \neq 0$ for some $j \geq 2$, the six squarefree cubics divisible by x_j and $x_1^2 x_j$, projected modulo $x_j L$, give seven independent quadrics. The pure x_1 direction has rank ten.

For $s = 2$, a basis consists of the seven squarefree cubics containing at most one of x_1, x_2 , the six cubics

$$x_2^2 x_j + 2x_1 x_2 x_j, \text{ quad } x_1 x_2 x_j + \frac{1}{2} x_1^2 x_j \quad (3 \leq j \leq 5),$$

and $x_1 x_2 (x_1 + x_2)$. If $a_j \neq 0$ for some $j \geq 3$, five squarefree inputs and the two special inputs give, modulo $x_j L$, five distinct pair monomials and the independent quadrics $x_2^2 + 2x_1 x_2$ and $x_1 x_2 + \frac{1}{2} x_1^2$. If $\lambda = a_1 \partial_{x_1} + a_2 \partial_{x_2}$, the image contains the three pair monomials on x_3, x_4, x_5 , a nonzero direction in each $\text{span}\{x_1 x_j, x_2 x_j\}$, and the nonzero quadratic

$$a_2 x_1^2 + 2(a_1 + a_2) x_1 x_2 + a_1 x_2^2.$$

Thus every nonzero directional map has rank at least seven, and every positive-rank quotient symbol has rank at least six after reducing by R .

For $d = 4$, let $S = \ker P$ be a line and let v lie in the symbol kernel. Its four complementary derivatives span a subspace of R . If that span were nonzero, the essential-variable space of v would have dimension at most two and would support a nonzero element of E_2 , contrary to the rank-four floor in Lemma 1.3. Hence $v \in \text{Sym}^3 S$, so the kernel has dimension at most one. The full symbol at $d = 5$ is injective. With (3.1), the $s = 1, 2$ row is

$$(4.9) \quad (3, 9, 9, 10, 16, 17).$$

Every entry of (4.4), (4.5), (4.8), and (4.9) is at least $10d/3$. This proves (4.1). $\square$

Remark 4.2 (The repaired boundary). The formal pair- and triple-product spans for T_2 have dimensions 12 and 17, while the actual derivative spaces have dimensions 11 and 14. For T_3 , the formal triple-product span has dimension 19 while the actual middle derivative space has dimension 18. The proof above deliberately uses the formal squarefree table only in the six-independent-factor case. The negative equalities are frozen as regression tests.

5. A SYMMETRIC IMAGE-SPAN INEQUALITY

Lemma 5.1. *Let $A_i : W^* \rightarrow W$ be symmetric maps, let $r_i = \text{rank } A_i$, and put $D = \dim \sum_i \text{im } A_i$. Then*

$$(5.1) \quad \text{rank} \left(\sum_i A_i \right) \geq 2D - \sum_i r_i.$$

Proof. Choose $B_i : \mathbb{K}^{r_i} \rightarrow W$ onto $\text{im } A_i$. Symmetry gives $\ker A_i = (\text{im } A_i)^\perp$, so $A_i = B_i J_i B_i^*$ for an invertible symmetric J_i . With $B = [B_1 \ \cdots \ B_N]$ and block-diagonal J , Sylvester's inequality gives

$$\text{rank}(B J B^*) \geq \text{rank}(B J) + \text{rank}(B^*) - \sum_i r_i = 2D - \sum_i r_i.$$

□

6. THE GLOBAL LOWER BOUND

Suppose

$$\text{perm}_6 = \sum_{i=1}^N T_i$$

with no zero summands. Let

$$A_i = C_{3,3}(T_i), \quad U_i = \text{im } A_i, \quad u_i = \dim U_i, \quad \delta_i = 20 - u_i, \quad \Delta = \sum_i \delta_i.$$

Put $U = \sum_i U_i$ and $\dim U = 400 + h$. The sum of the symmetric maps A_i is the permanent middle catalecticant, with rank 400 and image E_3 . Lemma 5.1 gives

$$(6.1) \quad \boxed{h \leq 10N - 200 - \frac{\Delta}{2}.$$

Let $F_i = \mathcal{D}_2(T_i)$, $F = \sum_i F_i$, and $Q = F/E_2$. Linearity of the derivative spaces gives $E_2 \subset F$. Blockwise differentiation modulo E_2 defines

$$\tilde{\beta} : \bigoplus_i U_i \longrightarrow V \otimes Q.$$

Its kernel consists precisely of tuples for which every first derivative of $\sum_i g_i$ lies in E_2 , equivalently $\sum_i g_i \in E_3$ by Lemma 1.3. Since $E_3 \subset U$,

$$(6.2) \quad \text{rank } \tilde{\beta} = \dim(U/E_3) = h.$$

Let L_i be the actual factor span of T_i . Since $E_2 \subset F$, $\partial E_2 = V$, and $\partial F_i \subset L_i$, the spaces L_i span all 36 variables. Choose any ordering and put

$$W_i = \sum_{j \leq i} L_j, \quad d_i = \dim(W_i/W_{i-1}), \quad \sum_i d_i = 36.$$

Let $Z_i \subset L_i \otimes Q$ be the image of the i -th block and let $H_i = \sum_{j \leq i} Z_j$. Projection

$$\pi_i : W_i \otimes Q \longrightarrow (W_i/W_{i-1}) \otimes Q$$

kills H_{i-1} , hence

$$(6.3) \quad \dim H_i - \dim H_{i-1} \geq \dim \pi_i(H_i) = \dim \pi_i(Z_i).$$

On the current block, the first-factor map is

$$P_i : L_i \twoheadrightarrow L_i/(L_i \cap W_{i-1}), \quad \text{rank } P_i = d_i.$$

The natural map

$$F_i/(F_i \cap E_2) \longrightarrow F/E_2$$

is injective. Therefore $\dim \pi_i(Z_i)$ is exactly the local symbol rank in Proposition 4.1. Summing (6.3) gives

$$(6.4) \quad h \geq \sum_i \left(\frac{10}{3} d_i - \frac{\delta_i}{2} \right) = 120 - \frac{\Delta}{2}.$$

Comparing (6.1) and (6.4) cancels the complete individual middle-rank defect and yields $N \geq 32$.

For the reverse inequality, Glynn's identity [1] gives

$$\text{perm}_6 = \frac{1}{32} \sum_{\epsilon_1=1, \epsilon_2, \dots, \epsilon_6=\pm 1} \left(\prod_{r=1}^6 \epsilon_r \right) \prod_{c=1}^6 \left(\sum_{r=1}^6 \epsilon_r x_{rc} \right).$$

Walsh cancellation leaves exactly the monomials that use each row once. This is a 32-term Chow decomposition. Theorem 1.2 follows.

7. EXACT REPLAY AND AUDIT BOUNDARY

From the repository checkout, run

```
python scripts/n6_exact_ordinary_chow_rank_32.py \
--verify-json data/n6_exact_ordinary_chow_rank_32.json
python -m unittest tests.test_n6_exact_ordinary_chow_rank_32 -v
```

The first command derives the coordinate intersection maxima, all 45,696 coordinate cases behind (3.2), the actual normal-form derivative ranks and squarefree subspaces in (4.7)–(4.8), the low-span monomial floors, and the final cancellation. It also records the formal/actual mismatches in Remark 4.2.

The Borel fixed-point reductions, prolongation identity, elementary directional arguments, symmetric image-span lemma, and filtration increment (6.3) are written proofs, not numerical inferences. This is an internal computation-assisted exact proof; named external peer review and proof-assistant formalization remain outstanding.

Remark 7.1 (Scope). The theorem concerns ordinary Chow rank over characteristic zero. It does not determine border Chow rank or prove $R_{\text{Chow}, \mathbb{K}}(\text{perm}_n) = 2^{n-1}$ for general n . Product rank and Chow rank are used synonymously in the relevant literature; see [2].

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